

Savíns Puertas Martín, Ph.D.

✉ savinspm@ual.es

🆔 <https://orcid.org/0000-0001-8956-1733>



Employment History

- 01.05.2025 – now 📌 **Post-doctoral Researcher** (funded by the Junta de Andalucía). Dept. of Informatics, University of Almería.
- 01.02.2022 – 31.12.2024 📌 **Post-doctoral Researcher** (Margarita Salas / Recualificación grant, funded by EU NextGenerationEU). 2-year research stay at the University of Sheffield (Chemoinformatics Research Group), followed by 11 months at the University of Almería.
- 01.02.2021 – 31.01.2022 📌 **Post-doctoral Researcher** working on the project *High Performance Computing for Optimizing Intensity Modulated Radiation Therapy Schedules*. Dept. of Informatics, University of Almería.
- 01.10.2016 – 30.09.2020 📌 **Pre-doctoral Researcher. National Competitive Scholarship FPU (Formación de Profesorado Universitario (University Teacher Training))**. Dept. of Informatics, University of Almería.
- 01.06.2016 – 30.09.2016 📌 **Research Labor Staff**. Supercomputing-Algorithm Research Group, Dept of Informatics, University of Almería.

Education

- 2016 – 2020 📌 **PhD in Informatics**, University of Almería, Spain.
Score: Cum laude (maximum score). International Mention.
Thesis title: *High Performance Computing for Optimization Problem Solving in Bioinformatics*.
- 2015 – 2017 📌 **Master in Informatics**, University of Almería, Spain.
Average score: 9.79 out 10.
Final master project title: *Solving optimization problems in the pharmaceutical industry through evolutionary algorithms*. Score: 10 out 10 with Honors.
- 2011 – 2015 📌 **Degree in Informatics**, University of Almería, Spain.
Average score: 9.55 out 10.
Final degree project title: *a voice interaction system for handling mashup interfaces*. Score: 10 out 10 with Honors.

Awards and Achievements

- 2021 – 2024 📌 **Margarita Salas Grant** (Recualificación del sistema universitario español), funded by the European Union NextGenerationEU. University of Almería. Duration: 3 years.
- 2022 📌 **Premio Jóvenes Investigadores Informáticos**, VI Edición Premios de Investigación Sociedad Científica Informática de España (SCIE) – Fundación BBVA. Award: Diploma and 5,000 Euros.
- 2021 📌 **Third Prize, “Tu Tesis en 3 minutos” Contest**, Asociación SARTECO. Award: Diploma and 300 Euros.

Awards and Achievements (continued)

- 2016 - 2020  **FPU Scholarship**, Spanish Ministry of Education, Culture and Sport. The FPU scholarship provides financial support for awardees to complete an official PhD program and to carry out research and teaching activities, in any discipline, by signing a predoctoral contract. Code: FPU15/02912.
- 2018  **Best final project award in Informatics Master**. EPS-FCCEE - University of Almería.
 **Best academic record award in Informatics Master**. University of Almería. Informatics Master.
- 2016  **Collaboration grant with the Department of Informatics of the University of Almería**, Spanish Ministry of Education, Culture and Sport.
Project Title: *Optimization of protein folding using HPC techniques*.
- 2015  **Best final project award in Informatics Degree**. EPS-FCCEE - University of Almería.
 **Best academic record award in Informatics Degree**. University of Almería. Informatics Degree.

Research Publications

Journal Articles

- [1] J. J. Moreno, **S. Puertas-Martín**, N. G. Roman, J. L. Redondo, and E. M. Garzón, "Fast Automatic Radiotherapy Planning via Algorithmic Improvements and Computational Acceleration," *Future Generation Computer Systems*, vol. 176, p. 108 168, Mar. 2026, ISSN: 0167-739X.  DOI: 10.1016/j.future.2025.107768.
- [2] J. J. Moreno, **S. Puertas-Martín**, J. L. Redondo, P. M. Ortigosa, and E. M. Garzón, "Where High-Performance Computing Meets Radiotherapy for Enhanced Intensity-Modulated Radiation Therapy Planning," *Concurrency and Computation: Practice and Experience*, vol. 37, May 2025, ISSN: 1532-0626.  DOI: 10.1002/cpe.8399.
- [3] J. J. Moreno, **S. Puertas-Martín**, J. L. Redondo, P. M. Ortigosa, A. Zawadzka, P. Kukołowicz, R. Szmurło, I. Kaliszewski, J. Miroforidis, and E. M. Garzón, "Bi-Level Optimization to Enhance Intensity Modulated Radiation Therapy Planning," *Informatica*, pp. 1–26, Sep. 2024, ISSN: 0868-4952.  DOI: 10.15388/24-INFOR560.
- [4] N. C. Cruz, **S. Puertas-Martín**, J. L. Redondo, and P. M. Ortigosa, "An Effective Solution for Drug Discovery Based on the Tangram Meta-Heuristic and Compound Filtering," *Informatica*, vol. 34, no. 4, pp. 743–769, Nov. 2023, ISSN: 0868-4952.
- [5] J. S. García, **S. Puertas-Martín**, J. L. Redondo, J. J. Moreno, and P. M. Ortigosa, "Improving drug discovery through parallelism," *The Journal of Supercomputing*, vol. 79, pp. 9538–9557, Jan. 2023, ISSN: 0920-8542.  DOI: 10.1007/s11227-022-04950-3.
- [6] M. R. Ferrández, **S. Puertas-Martín**, J. L. Redondo, H. Pérez-Sánchez, and P. M. Ortigosa, "A two-layer mono-objective algorithm based on guided optimization to reduce the computational cost in virtual screening," *Scientific Reports*, vol. 12, p. 12 769, 2022, ISSN: 2045-2322.  DOI: 10.1038/s41598-022-16913-w.
- [7] **S. Puertas-Martín**, J. L. Redondo, M. R. Ferrández, H. Pérez-Sánchez, and P. M. Ortigosa, "MultiPharm-DT: A Multi-Objective Decision Tool for Ligand-Based Virtual Screening Problems," *Informatica*, vol. 33, no. 1, pp. 55–80, 2021, ISSN: 0868-4952.
- [8] **S. Puertas-Martín**, J. L. Redondo, H. Pérez-Sánchez, and P. M. Ortigosa, "Optimizing Electrostatic Similarity for Virtual Screening: A New Methodology," *Informatica*, pp. 1–19, Jul. 2020, ISSN: 0868-4952.  DOI: 10.15388/20-INFOR424.  [Online]. Available: <https://informatica.vu.lt/doi/10.15388/20-INFOR424>.

- [9] **S. Puertas-Martín**, A. J. Banegas-Luna, M. Paredes-Ramos, J. L. Redondo, P. M. Ortigosa, O. O. Brovarets', and H. Pérez-Sánchez, "Is high performance computing a requirement for novel drug discovery and how will this impact academic efforts?" *Expert Opinion on Drug Discovery*, vol. 15, no. 9, pp. 981–986, 2020, ISSN: 1746045X.  DOI: 10.1080/17460441.2020.1758664.  [Online]. Available: <https://doi.org/10.1080/17460441.2020.1758664>.
- [10] **S. Puertas-Martín**, J. L. Redondo, P. M. Ortigosa, and H. Pérez-Sánchez, "OptiPharm: An evolutionary algorithm to compare shape similarity," *Scientific Reports*, vol. 9, no. 1, p. 1398, Dec. 2019, ISSN: 2045-2322.  DOI: 10.1038/s41598-018-37908-6.  [Online]. Available: <https://doi.org/10.1038/s41598-018-37908-6>.
- [11] A. J. Banegas-Luna, J. P. Cerón-Carrasco, **S. Puertas-Martín**, and H. Pérez-Sánchez, "BRUSELAS: HPC Generic and Customizable Software Architecture for 3D Ligand-Based Virtual Screening of Large Molecular Databases," *Journal of Chemical Information and Modeling*, vol. 59, no. 6, pp. 2805–2817, Jun. 2019, ISSN: 1549-9596.  DOI: 10.1021/acs.jcim.9b00279.  [Online]. Available: <https://pubs.acs.org/doi/10.1021/acs.jcim.9b00279>.
- [12] M. R. Ferrández, **S. Puertas-Martín**, J. L. Redondo, B. Ivorra, A. M. Ramos, and P. M. Ortigosa, "High-performance computing for the optimization of high-pressure thermal treatments in food industry," *The Journal of Supercomputing*, vol. 75, no. 3, pp. 1187–1202, Mar. 2019, ISSN: 0920-8542.  DOI: 10.1007/s11227-018-2351-4.  [Online]. Available: <http://link.springer.com/10.1007/s11227-018-2351-4>.

International Conferences

- [1] **S. Puertas-Martín**, J. J. Moreno, J. L. Redondo, P. M. Ortigosa, and E. M. Garzón, "Efficient load scheduling of imrt planning in heterogeneous multicore clusters," in *Proceedings of the International Conference on Parallel Processing and Applied Mathematics (PPAM 2024)*, Technical University of Ostrava, Ostrava, Czech Republic, Nov. 2024.
- [2] **S. Puertas-Martín**, J. L. Redondo, P. M. Ortigosa, and V. Gillet, "Improving partial matching accuracy precision with evolutionary algorithm in shape ligand based virtual screening," in *24th European Symposium on Quantitative Structure-Activity Relationship (EuroQSAR 2024)*, Chemoinformatics and Modeling Society, Barcelona, Spain, Sep. 2024.
- [3] J. J. Moreno, **S. Puertas-Martín**, J. L. Redondo, P. M. Ortigosa, and E. M. Garzón, "Exploiting multicore servers to optimize imrt radiotherapy planning," in *EuroPAR'24. 1st Workshop about High-Performance e-Science*, Universidad Carlos III de Madrid, Madrid, Spain, Aug. 2024.
- [4] N. C. Cruz, **S. Puertas-Martín**, J. L. Redondo, and P. M. Ortigosa, "Preliminary results of using the tangram meta-heuristic for virtual screening in drug discovery," in *11th International Work-Conference on Bioinformatics and Biomedical Engineering (IWBBIO 2023)*, University of Granada, Gran Canaria, Spain, Jul. 2023.
- [5] J. J. Moreno, **S. Puertas-Martín**, P. M. Ortigosa, J. L. Redondo, and E. M. Garzón, "Imrt planning based on the geud model and evolutionary optimization on multicore systems," in *Proceedings of the 23th International Conference Computational and Mathematical Methods in Science and Engineering (CMMSE 2023)*, Universidad de Cádiz, Rota, Spain, Jul. 2023.
- [6] **S. Puertas-Martín** and V. Gillet, "Partial matching shape and coloured-based applied to ligand-based virtual screening," in *Ninth Joint Sheffield Conference on Chemoinformatics*, University of Sheffield, Sheffield, UK, Jun. 2023.
- [7] **S. Puertas-Martín**, J. L. Redondo, H. Pérez-Sánchez, V. J. Gillet, and P. M. Ortigosa, "Optimization based on atom-pair properties in flexible ligands by optipharma," in *XV. Workshop on Global Optimization*, University of Szeged, Szeged, Hungary, Sep. 2022.
- [8] J. S. García, **S. Puertas-Martín**, J. L. Redondo, H. Pérez-Sánchez, and P. M. Ortigosa, "Optipharma-parallel: Enhancing the virtual screening process via parallelism," in *Proceedings of the 22th International*

Conference Computational and Mathematical Methods in Science and Engineering (CMMSE 2022), Universidad de Salamanca, Rota, Spain, Jul. 2022.

- [9] J. J. Moreno, **S. Puertas-Martín**, J. L. Redondo, L. González Casado, P. M. Ortigosa, and E. M. Garzón, "A new hybrid optimization algorithm to combine physical and biological criteria to compute imrt planning," in *19th Workshop on Advances in Continuous Optimization (EurOpt 2022)*, Universidade Nova de Lisboa, Lisbon, Portugal, Jul. 2022.
- [10] J. J. Moreno, **S. Puertas-Martín**, J. L. Redondo, P. M. Ortigosa, and E. M. Garzón, "Acceleration of the tuning of eud model for imrt planning on multigpu clusters," in *Proceedings of the 22th International Conference Computational and Mathematical Methods in Science and Engineering (CMMSE 2022)*, Universidad de Salamanca, Rota, Spain, Jul. 2022.
- [11] **S. Puertas-Martín**, J. L. Redondo, A. J. Banegas-Luna, E. M. Garzón, H. Pérez-Sánchez, V. J. Gillet, and P. M. Ortigosa, "Virtual screening based on electrostatic similarity and flexible ligands," in *22nd International Conference on Computational Science and Its Applications (ICCSA 2022)*, Universidad de Málaga, Malaga, Spain, Jul. 2022.
- [12] **S. Puertas-Martín**, J. L. Redondo, V. J. Gillet, H. Pérez-Sánchez, and P. M. Ortigosa, "Optimization method for maximizing the similarity in flexible molecules," in *19th Workshop on Advances in Continuous Optimization (EurOpt 2022)*, Universidade Nova de Lisboa, Lisbon, Portugal, Jul. 2022.
- [13] **S. Puertas-Martín**, J. J. Moreno, J. L. Redondo, P. M. Ortigosa, A. Zawadzka, J. Miroforidis, I. Kaliszewski, and E. M. Garzón, "New multiobjective strategies to optimize constrained eud radiotherapy models," in *26th International Conference on Multiple Criteria Decision Making (MCDM 2022)*, University of Portsmouth, Portsmouth, UK, Jun. 2022.
- [14] **S. Puertas-Martín**, J. L. Redondo, E. M. Garzón, H. Pérez-Sánchez, and P. M. Ortigosa, "Increasing the accuracy of optipharm's virtual screening predictions by implementing molecular flexibility," in *9th International Work-Conference on Bioinformatics and Biomedical Engineering (IWBBIO 2022)*, Universidad de Granada, Gran Canaria, Spain, Jun. 2022.
- [15] **S. Puertas-Martín**, J. J. Moreno, J. L. Redondo, P. M. Ortigosa, and E. M. Garzón, "Optimizing eud model parameters in radiotherapy planning," in *The 7th Int'l Conf on Health Informatics and Medical Systems*, American Council on Science and Education, Las Vegas, USA, Jul. 2021.
- [16] **S. Puertas-Martín**, J. L. Redondo, H. Pérez-Sánchez, and P. M. Ortigosa, "Optimization of molecular descriptors using memetic algorithms," in *18th Workshop on Advances in Continuous Optimization, EurOpt2021*, Ecole Nationale de l'Aviation Civile, Toulouse, France, Jul. 2021.
- [17] **S. Puertas-Martín**, J. L. Redondo, H. Pérez-Sánchez, and P. M. Ortigosa, "Maximizing the electrostatic similarity in drug discovery through evolutionary algorithms," in *8th International Work-Conference on Bioinformatics and Biomedical Engineering*, University of Granada, Granada, Spain, Sep. 2020.
- [18] **S. Puertas-Martín**, J. L. Redondo, H. Pérez-Sánchez, and P. M. Ortigosa, "Virtual screening in electrostatic potential using an evolutionary algorithm," in *Proceedings of META18: 7th International Conference on Metaheuristics and Nature Inspired Computing*, Université de Lille, Marrakech, Morocco, Oct. 2018, pp. 207–209.
- [19] **S. Puertas-Martín**, J. L. Redondo, H. Pérez-Sánchez, and P. M. Ortigosa, "Multi-objective evolutionary algorithm for evaluation of shape and electrostatic similarity," in *Proceedings of LeGO 2018 - Int. Workshop on Global Optimization. AIP Conference Proceedings*, Leiden University, vol. 1, Sep. 2018, pp. 1–4, ISBN: 9780735417984. DOI: 10.1063/1.5089986. [Online]. Available: <http://aip.scitation.org/doi/abs/10.1063/1.5089986>.
- [20] **S. Puertas-Martín**, J. L. Redondo, H. Pérez-Sánchez, and P. M. Ortigosa, "Optimizing electrostatic similarity using a global evolutionary algorithm," in *Proceedings of 6th EUROPT Workshop on Advances in Continuous Optimization*, University of Almería, Almería, Spain, Jul. 2018.

- [21] **S. Puertas-Martín**, J. L. Redondo, H. Pérez-Sánchez, and P. M. Ortigosa, "Virtual screening in molecular shape by using an evolutionary algorithm," in *Proceedings OLA'2018 International Workshop on Optimization and Learning: Challenges and Applications*, Université de Lille, Alicante, Spain, Feb. 2018, pp. 63–64.
- [22] M. R. Ferrández, **S. Puertas-Martín**, J. L. Redondo, B. Ivorra, R. A. M., and O. P. M., "High-performance computing for optimizing high-pressure thermal treatments in food processing," in *Proceedings of the 17th International Conference on Computational and Mathematical Methods in Science and Engineering, CMMSE 2017*, University of Cadiz, Alicante, Spain, Jul. 2017, pp. 862–869, ISBN: 978-84-617-8694-7.
- [23] **S. Puertas-Martín**, M. R. Ferrández, J. L. Redondo, H. Pérez-Sánchez, and P. M. Ortigosa, "Enhancing molecular shape comparison by a parallel global evolutionary algorithm," in *Proceedings of the 17th International Conference on Computational and Mathematical Methods in Science and Engineering, CMMSE 2017*, University of Cadiz, Rota, Spain, Jul. 2017, pp. 1722–1728, ISBN: 978-84-617-8694-7.
- [24] **S. Puertas-Martín**, M. R. Ferrández, J. L. Redondo, H. Pérez-Sánchez, and P. M. Ortigosa, "Enhancing molecular shape comparison by a parallel global evolutionary algorithm," in *Proceedings of the 17th International Conference on Computational and Mathematical Methods in Science and Engineering, CMMSE 2017*, University of Cadiz, Rota, Spain, Jul. 2017, pp. 1722–1728, ISBN: 978-84-617-8694-7.
- [25] **S. Puertas-Martín**, H. den-Haan, J. L. Redondo, H. Pérez-Sánchez, and P. M. Ortigosa, "Enhancing molecular shape comparison by a global evolutionary algorithm," in *4th International Work-Conference on Bioinformatics and Biomedical Engineering*, University of Granada, Granada, Spain, 2016.
- [26] **S. Puertas-Martín**, J. L. Redondo, H. den-Haan, H. Pérez-Sánchez, and P. M. Ortigosa, "Multi-objective based scoring function for ligand based virtual screening," in *Proceedings of the XIII Global Optimization Workshop*, University of Minho, Braga, Portugal, 2016, ISBN: 978-989-20-6764-3.

Education Conferences

- [1] J. J. Moreno, **S. Puertas-Martín**, F. Orts, N. C. Cruz, J. L. Redondo, E. Garzón, and O. P. M., "On simulating an arm processor for teaching computer structure," in *12th annual International Conference of Education, Research and Innovation (ICERI 2019)*, Seville, Spain, Dec. 2019.
- [2] F. Orts, N. C. Cruz, **S. Puertas-Martín**, M. Ruiz-Ferrández, J. J. Moreno, C. Medina-López, P. M. Ortigosa, V. Ruíz, L. Casado, J. M. Salmerón, J. L. Redondo, G. E. Garzón, G. Ortega, and R. Villegas, "Learning quantum computation through simple examples," in *11th International Conference of Education, Research and Innovation (ICERI 2019)*, Seville, Spain, Dec. 2019.
- [3] F. Orts, N. C. Cruz, **S. Puertas-Martín**, M. Ruiz-Ferrández, J. J. Moreno, C. Medina-López, P. M. Ortigosa, V. Ruíz, L. Casado, J. M. Salmerón, J. L. Redondo, G. E. Garzón, G. Ortega, and R. Villegas, "Learning quantum computation through simple examples," in *11th International Conference of Education, Research and Innovation (ICERI 2019)*, Seville, Spain, Dec. 2019.
- [4] N. C. Cruz, M. R. Ferrández, E. M. Garzón, J. M. García Salmerón, L. González Casado, V. González Ruiz, C. Medina-López, J. J. Moreno, G. Ortega López, P. M. Ortigosa, F. J. Orts, **S. Puertas-Martín**, J. L. Redondo, J. F. Sanjuan Estrada, and T. Santamaría López, "Simulación de un procesador arm para la enseñanza de estructura de computadores. poster," in *Jornadas de Innovación Docente y Experiencias Profesionales en la Universidad de Almería. Curso 2019-20*, Vicerrectorado de Enseñanzas Oficiales y Formación Continua y por la Coordinación de Gobierno e Interacción con la Sociedad y las Empresas de la Universidad de Almería, Almería, Spain, Sep. 2019.
- [5] **S. Puertas-Martín**, J. J. Moreno, F. J. Orts, N. C. Cruz, J. L. Redondo, E. M. Garzón, and P. M. Ortigosa, "Simulación de un procesador arm para la enseñanza de estructura de computadores," in *Actas de las Jornadas SARTECO 2019*, Cáceres, Spain, Sep. 2019, pp. 235–240, ISBN: 978-84-09-12127-4.
- [6] V. González-Ruiz, G. Ortega, E. Garzón, N. Calvo-Cruz, R. J. L., J. Salmerón, L. Casado, P. M. Ortigosa, C. Medina-López, J. J. Moreno, M. R. Ferrández, F. Orts, **S. Puertas-Martín**, and T. Santamaría-López, "Project-based learning: An experience," in *11th International Conference on Education and New Learning Technologies (EDULEARN 2019)*, Palma, Spain, Jul. 2019.

- [7] N. C. Cruz, M. R. Ferrández, E. M. Garzón, J. M. García Salmerón, L. González Casado, V. González Ruiz, C. Medina-López, J. J. Moreno, G. Ortega López, P. M. Ortigosa, F. J. Orts, **S. Puertas-Martín**, J. L. Redondo, J. F. Sanjuan Estrada, T. Santamaría López, and R. Villegas, "Recursos para el aprendizaje de computación cuántica en el grado de ingeniería informática. poster," in *Jornadas de Innovación Docente y Experiencias Profesionales en la Universidad de Almería. Curso 2018-19*, Vicerrectorado de Enseñanzas Oficiales y Formación Continua y por la Coordinación de Gobierno e Interacción con la Sociedad y las Empresas de la Universidad de Almería, Almería, Spain, Sep. 2018.

National Conferences

- [1] **S. Puertas-Martín**, J. J. Moreno Riado, and V. González-Ruiz, "Un intercomunicador de audio de baja latencia con compresión perceptual basada en paquetes wavelet para la docencia de tecnologías multimedia," in *Actas de las XXXVI Jornadas SARTECO 2026*, Madrid, Spain, Jun. 2026.
- [2] M. A. Sánchez-Olmedo, **S. Puertas-Martín**, A. de Torres Sánchez, P. Mezquita-Raya, and J. L. Redondo, "Infraestructura p2p agnóstica para streaming multimedia seguro en entornos clínicos: Soberanía del dato mediante virtualización ia on-premise," in *Actas de las XXXVI Jornadas SARTECO 2026*, Madrid, Spain, Jun. 2026.
- [3] J. J. Moreno Riado, **S. Puertas-Martín**, J. L. Redondo, P. M. Ortigosa, and E. M. Garzón, "Clústeres heterogéneos para acelerar la planificación automática en radioterapia," in *Actas de las XXXV Jornadas SARTECO 2025*, Sevilla, Spain, Jun. 2025.
- [4] J. J. Moreno, **S. Puertas-Martín**, M. Padilla, J. L. Redondo, P. M. Ortigosa, and E. M. Garzón, "Aceleración del ajuste automático y personalizado de modelos geud para planificación en imrt," in *Actas de las Jornadas SarteCO CEDI 2024*, Universidad de La Coruña, A Coruña, Spain, Jun. 2024.
- [5] M. R. Ferrández, **S. Puertas-Martín**, J. L. Redondo, B. Ivorra, A. M. Ramos, and P. M. Ortigosa, "Computación de alto rendimiento para optimizar tratamientos térmicos de alta presión en la industria alimenticia," in *Avances en arquitectura y tecnología de computadores. Actas de las Jornadas SARTECO 2017*, Málaga, Spain, Sep. 2017, pp. 119–122, ISBN: 978-84-697-4835-0.
- [6] **S. Puertas-Martín**, M. R. Ferrández, J. L. Redondo, H. Pérez-Sánchez, and P. M. Ortigosa, "Cribado virtual mediante un algoritmo evolutivo global paralelo," in *Actas de las Jornadas SARTECO 2017*, Cácares, Spain, Sep. 2017, pp. 177–179, ISBN: 978-84-697-4835-0.
- [7] **S. Puertas-Martín**, M. R. Ferrández, J. L. Redondo, H. Pérez-Sánchez, and P. M. Ortigosa, "Algoritmo evolutivo global como herramienta de cribado virtual utilizando la forma molecular," in *III Jornadas Doctorales de la Universidad de Murcia*, Murcia, Spain, May 2017, pp. 236–240, ISBN: 978-84-608-9779-8.

Technology transfer, Patents and Intellectual Properties

- | | |
|-------------------------|--|
| 2018 |  OPTIPHARM: an innovative evolutionary algorithm for virtual screening.
P. M. Ortigosa, J. L. Redondo, S. Puertas-Martín and H. Pérez-Sánchez.
University of Almería and Universidad Católica San Antonio de Murcia.
Application Number: 201899900606752. Register number: RTA-94-18.
Registration Date: 16/02/2018. Concession date: 09/08/2018. |
| 01.04.2016 – 30.06.2016 |  Evaluation of high-performance computing infrastructures.
Contract.
Number of researchers: 14.
Funding: BULL ESPANA, S.A.
Total amount: 6.109,83 Euros. |

Research Stays

- Feb. 2022 – Jan. 2024  **Postdoctoral Stay**, Chemoinformatics Research Group, Information School, University of Sheffield.
Sheffield, S1 4DP, United Kingdom.
- Sep. 2018 – Dec. 2018  **Predoctoral Stay**, Centre for Logistics and Heuristic Optimization (CLHO), Kent Business School, University of Kent.
Canterbury, CT2 7NZ, United Kingdom.

Projects

- 01.09.2025 – 30.08.2028  **Soluciones Computacionales en el Campo de la Salud (Comp4Health-II).**
National project.
Principal Research: Juana López Redondo and Gracia Ester Martín Garzón.
Number of researchers: 32.
Total amount: 363.250 Euros.
- 01.09.2022 – 30.09.2025  **Diseño de métodos computacionales con aplicación en salud (Comp4Health).**
National project.
Principal Research: Gracia Ester Martín Garzón and Juana López Redondo.
Number of researchers: 32.
Total amount: 227.177 Euros.
- 01.12.2022 – 30.11.2024  **Pulsera inteligente para la predicción, detección y notificación de ataques epilépticos (EPILSERA).**
Principal Research: Pilar Martínez Ortigosa and Gracia Ester Martín Garzón.
Number of researchers: 21.
Funding: Ministerio de Ciencia e Innovación.
Total amount: 143.750 Euros.
- 01.01.2020 – 30.06.2023  **Desarrollo de librerías cuánticas eficientes (DELICE).**
Principal Research: Gloria Ortega and Juan Francisco Sanjuan.
Number of researchers: 12.
Total amount: 29.914 Euros.
- 01.01.2020 – 31.12.2022  **Inteligencia Computacional en descubrimiento de fármacos (IC-DRUGS).**
Principal Research: Pilar Martínez Ortigosa and Juana López Redondo.
Number of researchers: 11.
National project.
Principal Research: Pilar Martínez Ortigosa and Gracia Ester Martín Garzón.
Number of researchers: 31.
Code: RTI2018-095993-B-100.
Funding: Spanish Ministry of Science and Innovation. National Program for Scientific Research, Development and Technological Innovation.
Total amount: 186.461,00 Euros.

Projects (continued)

- 15.10.2019 – 14.10.2021  **Computación de Altas Prestaciones para Optimizar Planificaciones de Radioterapia de Intensidad Modulada (High Performance Computing to Optimize Intensity Modulated Radiation Therapy Planning).**
Regional project.
Principal Research: Ester Martín Garzón and Juana López Redondo.
Number of researchers: 14.
Code: UAL18-TIC-Ao2o-B.
Funding: Junta de Andalucía, Fondos Feder UAL.
Total amount: 76.800,00 Euros.
- 01.01.2019 – 31.12.2019  **Descubrimiento y optimización de compuestos bioactivos mediante técnicas avanzadas de química computacional (Discovery and optimization of bioactive compounds through advanced computational chemistry techniques).**
Principal Research: Horacio Emilio Pérez Sánchez.
Number of researchers: 12.
Code: 20988/PI/18.
Funding: Fundación Séneca.
Total amount: 62.000 Euros.
- 01.01.2016 – 31.12.2019  **Metodologías computacionales para desafíos de la sociedad (Computational methodologies for societal challenges).**
National project.
Principal Research: Leocadio González Casado and Pilar Martínez Ortigosa.
Number of researchers: 21.
Code: TIN2015-66680-C2-1-R.
Funding: Spanish Ministry of Economy and Competitiveness.
Total amount: 147.620 Euros.

Teaching Innovation Projects

- 2019  **Desarrollo de Recursos para motivar el estudio de la Ingeniería de Computadores (Development of Resources to motivate the study of Computer Engineering).**
Convocatoria de Grupos Docentes para la Creación de Materiales Didácticos en la Universidad de Almería. Bienio 2018 y 2019 (Call for Teaching Groups for the Creation of Didactic Materials at the University of Almeria. Biennial 2018 and 2019).
Funding: University of Almería.
Total amount: 800 Euros.
- 2018  **Desarrollo de Recursos para motivar el estudio de la Ingeniería de Computadores (Development of Resources to motivate the study of Computer Engineering).**
Convocatoria de Grupos Docentes para la Creación de Materiales Didácticos en la Universidad de Almería. Bienio 2018 y 2019 (Call for Teaching Groups for the Creation of Didactic Materials at the University of Almeria. Biennial 2018 and 2019).
Funding: University of Almería.
Total amount: 800 Euros.

Teaching

- 2024 – 2025  **Degree in Informatics**, University of Almería, Spain.
Multiprocessors. Third year. 41 hours.
Multimedia technologies. Third year. 19 hours.
- 2023 – 2024  **Degree in Informatics**, University of Almería, Spain.
Peripherals and interfaces. Third year. 57 hours (English).
- 2019 – 2020  **Degree in Informatics**, University of Almería, Spain.
Structure and computer technology. First year. 48 hours.
Peripherals and interfaces. Third year. 19 hours.
- 2018 – 2019  **Degree in Informatics**, University of Almería, Spain.
Network access technologies. Third year. 50 hours.
Structure and computer technology. First year. 2 hours.
- 2017 – 2018  **Degree in Informatics**, University of Almería, Spain.
Peripherals and interfaces. Third year. 38 hours.
Network access technologies. Third year. 19 hours.
Multi-processors. Third year. 18 hours.

Final Project Supervisor

- 2024  **Degree in Informatics**, University of Almería, Spain.
Co-supervisor: Monitorización y Gestión de una Red de Gran Tamaño con un Controlador SDN. Javier Cholbi López. Grade: Notable.
Co-supervisor: Optimización Evolutiva y Computación Avanzada para Mejorar la Planificación de Tratamientos de Radioterapia. María López Padilla. Grade: Sobresaliente.
- 2023  **Master in Computer Engineering**, University of Sheffield, UK.
Co-supervisor: A Model with the Optimal Voxel Size to Create 3D Point Cloud for Molecules in Similarity Searching by Using Machine Learning. Hui Yu. Grade: Distinction.
- 2021  **Degree in Informatics**, University of Almería, Spain.
Co-supervisor: Monitorización Mediante Control Gestual. Antonio Ayala Tirado. Grade: Notable.
Co-supervisor: Reconocimiento de Caídas Con Visión Artificial. Raquel Gutierrez García. Grade: Sobresaliente.
- 2019 – 2020  **Degree in Informatics**, University of Almería, Spain.
Co-supervisor: SMARTFRIDGE: Recognition of Actions in a Refrigerator and Decision Making in a Smart Environment. Rafael Alejandro García Rodríguez. Grade: Matrícula de Honor.

Skills

Languages  C, C++, JAVA, Python, $\text{\LaTeX}$.
Parallel Programming

Skills (continued)

Paradigms	■ MPI, PThreads.
Software applied to drug discovery	■ OpenEye's Software (ROCS, EON, VIDA, OMEGA), OpenEye's Toolkit, Py- mol, Reflex3D, WEGA, OptiPharm.